Artificial Intelligence Laboratory End Sem Exam, Dt. 17 and 18 -May-2026, Program Definitions

Focal Points :

1. Hosting of Model is must for all the definitions in Flask, Django, Fast-API, Spring, NodeJS, etc. only with proper user-friendly interface.
2. Go for maximum number of model preparation for better accuracy.
3. A core problem definition model must not be pretrained except if mentioned in question below. Otherwise, it must be your wholly trained own model.
4. For viva one must show a step-by-step breakdown of your design and implemented architecture, framework and algorithm. Failure to explain the nodal milestones means one has not understood the problem and simply used AI generated solution.
5. During the viva, one must be able to explain the underlying concept used. Eg. if you have resolved issue using R-CNN and VGG-NET then a group must have indepth understanding of R-CNN and VGGNET, its working and architecture with justification as to why any synonymous technology is not used.
6. Marking is RELATIVE.
7. Viva and Presentation will start at 2:00pm 18-May-2026 in 207, 203b and 203a lab. All the groups for viva will come one by one as per the sequence.

PROBLEM DEFINITIONS

1. Develop a English to Hindi transliterator.
[Please mention source of dataset used to develop your model]

INPUT:

Narendra Modi Bharat ke pradhanmantri hain. Dilli mein India Gate sthit hai. Reliance Industries ek badi company hai. Aaj 10 May ko mausam accha hai. Yeh ghatna 2023 mein hui thi. . Unhone ₹500 daan kiye. Bharat ne 2020 mein lockdown lagaya tha. Swatantrata Diwas 15 August ko manaya jata hai. Mere paas Samsung Galaxy phone hai.

OUTPUT:

नरेंद्र मोदी भारत के प्रधानमंत्री हैं। दिल्ली में इंडिया गेट स्थित है। रिलायंस इंडस्ट्रीज एक बड़ी कंपनी है। आज 10 मई को मौसम अच्छा है। यह घटना 2023 में हुई थी। उन्होंने ₹500 दान किए। भारत ने 2020 में लॉकडाउन लगाया था। स्वतंत्रता दिवस 15 अगस्त को मनाया जाता है। मेरे पास सैमसंग गैलेक्सी फोन है।

2. Design an AI based Career counsellor which is highly intelligent and adaptive to the the answers given by student who wants career guidance to choose Engineering (Civil, Mechanical, Electrical, Computer Science), Science (Biology, Chemistry, Physics), BBA or MBA (Management).

Eg. Imagine a student how have cleared 12th standard with General or Science stream approaches university for admission. The University possess an intelligent chat bot which ask questions to students and decides students interest and suggest the stream. The bot should not ask direct questions but indirect questions pertaining to the field, it then analyze the answer and suggest the field. The question must not be direct but indirect extracting valuable information from student and know the interest.

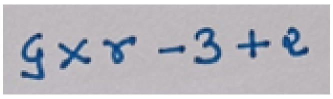
Design and implement an **Intelligent Career Counsellor Chatbot** for university admissions. Unlike traditional rule-based bots, this system must act as a human interviewer, using indirect questioning to uncover a student's latent interests and technical aptitude. The system should analyze open-ended responses to suggest a path in **Engineering** (Civil, Mech, Elec, CSE), **Pure Sciences** (Bio, Chem, Phys), or **Management** (BBA/MBA).

- **Adaptive Contextual Questioning:** The chatbot must not ask direct questions. Instead, it should use indirect inquiries. Based on the student's answer, the next question must pivot to dive deeper into that specific context.

[Note : Here you are allowed to use any pre-trained model. You may use any available dataset]

3. Gujarati Handwritten Mathematical equation digitization and solution.

Design and implement a basic handwritten mathematical equation written in Gujarati language to digitize and calculating the answer. Please find the image below for your reference. Prepare, collect, or scrap a dataset of handwritten Gujarati numerals (૦-૯) and operators (+, -, x, ÷).

Input: 

(An image or snip taken through device camera.

Output : ૬ x ૪ - ૩ + ૨ = ૨૩

4. Local vegetable variety classifier

Develop an end-to-end AI application that can accurately classify 5 common vegetables.

1. Ivy Gourd (Tindoda)
2. Pointed Gourd (Parvad)
3. Green Chilli (Lila Marcha)
4. Okra / Ladies Finger (Bhinda)
5. Peas (lila Vatana)

The model must be robust enough to differentiate between vegetables with similar morphological features. The system must be tested with real object and photographic images.

Eg.

Input :
Upload an Image or Snap through your device camera

Output : Pointed Gourd (Parvad)

Input :
Upload an Image or Snap through your device camera

Output : Okra / Ladies Finger (Bhinda)

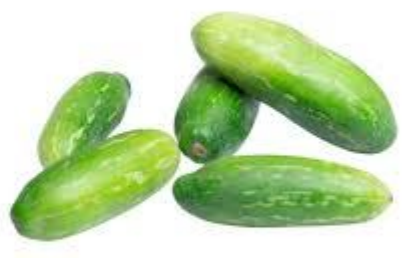
Input :
Upload an Image or Snap through your device camera

Output : Peas (Vatana)

Input :
Upload an Image or Snap through your device camera

Output : Green Chilli (Marcha)

Input :
Upload an Image or Snap through your device camera



Output : Ivy Gourd (Tindoda)

5. A software company wants to build automated resume ranking framework based on their job description to make the recruitment process easier, faster and more

efficient. The company receives resume through email or through web forms, which allows upload of resume in pdf or word format. The resume is format free. The candidate is free to use their own format for the resume building. The intelligent system looks at resume one by one and ranks it according to the job description.
[Note : Here you are allowed to use any pre-trained model. You may use any available dataset]